Kanoa Pick

Contact Information

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CV Last Updated September 5, 2026

Education

Ph.D. Expected Jun. 2029
Applied Ocean Science, Scripps Institution of Oceanography
Advisor: Falk Feddersen

M.S. Dec. 2025
Oceanography, Scripps Institution of Oceanography
Advisor: Falk Feddersen

B.S. Jun. 2024
Oceanic and Atmospheric Sciences, University of California San Diego
Advisor: Falk Feddersen
Minor: Engineering Mechanics

Professional Experience

Graduate Student Researcher Sep. 2024 – Present
Feddersen Group
Scripps Institution of Oceanography, University of California San Diego

Undergraduate Research Assistant Nov. 2021 – Sep. 2024
Feddersen Group
Scripps Institution of Oceanography, University of California San Diego

Journal Publications

Peer-Reviewed Journal Articles (Published & Accepted)

2. **Pick, K.** & Feddersen, F. (2026). Scaling the shape of shoaling and overturning solitary waves. *Journal of Fluid Mechanics*, 1040 (A8). https://doi.org/10.1017/jfm.2026.11869

1. Feddersen, F., Amador, A., **Pick, K.**, Vizuet, A., Quinn, K., Wolfinger, E., … Fincham, A. (2023). The Wavedrifter: A low-cost IMU-based Lagrangian drifter to observe steepening and overturning of surface gravity waves and the transition to turbulence. *Coastal Engineering Journal*, 66(1), 44–57. https://doi.org/10.1080/21664250.2023.2238949

Peer-Reviewed Journal Articles (In Revision)

1. **Pick, K.**, Quinn, K., Cirrito, N., Domi, A., Tsampras, G., Feddersen, F. (2026). BathyBoard: A surfboard-based platform for measuring surf zone bathymetry. *Journal of Surveying Engineering* (in revision).

Honors & Awards

National Defense Science and Engineering Graduate (NDSEG) Fellowship 2026 – 2029

Best Oral Presentation, Young Coastal Scientists and Engineers Conference–Americas — 2025

Best Paper Award, *Coastal Engineering Journal* — 2024

Shores Scholarship, *University of California San Diego* — 2021 – 2024

Presentations

Conference Oral Presentations & Research Seminars

8. Cader, J., Ismail, R., Halualani, G., Aduja, A., Borras, M., Salem, M., Viloria-Winnett, A., **Pick, K.**, Cader, A., Bickart, K. (2026). Neurological and Symptomatic Responses to Big Wave Surfing: A Prospective Pilot Study of Concussion Threshold-Crossing Events and Ocean Exposure Predictors. *Meeting of the International NeuroTrauma Society*, Oct. 25–29.
7. Cirrito, N., **Pick, K.**, Domi, A., Mack, C., Dooley, C., Young, D., Fiedler, J., O'Reilly, W., Young, A., Merrifield, M. (2026). A tower you can tow: automated beach morphology and wave runup from low-cost LiDAR. *Young Coastal Scientists and Engineers Conference–Americas*, Aug. 5–8.
6. **Pick, K.**, Feddersen, F., Tsampras, G. (2026). Effect of reef structures on wave breaking and overturning parameters. *International Conference on Coastal Engineering*, May 17–22.
5. **Pick, K.**, Feddersen, F., Tsampras, G. (2026). The effect of small engineered reefs on wave surfability. *Applied Ocean Sciences Seminar, Scripps Institution of Oceanography*, May 14.
4. **Pick, K.**, Feddersen, F., Tsampras, G. (2026). The effect of engineered reefs on wave breaking. *Ocean Sciences Meeting*, Feb. 22–27.
3. **Pick, K.** & Feddersen, F. (2025). Solitary wave overturning: Impact of beach slope and initial wave height. *Young Coastal Scientists and Engineers Conference–Americas*, Jul. 22–24.
2. **Pick, K.** & Feddersen, F. (2025). Solitary wave overturning: Impact of beach slope and initial wave height. *Waves In the Sea Environment Meeting*, May 12–15.
1. **Pick, K.** & Feddersen, F. (2025). Solitary wave overturning: Impact of beach slope and initial wave height. *Applied Ocean Sciences Seminar, Scripps Institution of Oceanography*, May 8.

Conference Poster Presentations

2. **Pick, K.** & Feddersen, F. (2025). The effect of tide and reef position on wave overturn shape: Preliminary steps. *Scripps Student Symposium, Scripps Institution of Oceanography*. Sep. 23.
1. **Pick, K.** & Feddersen, F. (2024). Overturning characteristics of solitary waves: The effects of beach slope and incident wave height. *Scripps Student Symposium, Scripps Institution of Oceanography*. Sep. 18.

Professional Development Workshops

National Hazards Engineering Research Infrastructure (NHERI) Computational Symposium — Feb. 5-7, 2025

Mentorship

Charlie Denham, Undergraduate Researcher — 2026 – Present
Scripps Institution of Oceanography
Mentoring an undergraduate research project using fully nonlinear potential flow

wave modeling to investigate how quadratic bathymetric profiles influence the shape of overturning waves. Providing guidance in numerical modeling, study design, data analysis, and scientific communication.

Media

Research covered by Environment Coastal & Offshore Magazine — Sep. 2026

Marine-industry coverage emphasizing implications for wave breaking and coastal sediment transport.

Research syndicated by Phys.org — Sep. 2026

International science-news coverage of my JFM study's predictions of breaking wave shape.

Research profile in University of California San Diego Today — Sep. 2026

Profile connecting my JFM research on breaking wave geometry with my experience as a surfer.

Scripps Institution of Oceanography science communication video and article — Sep. 2026

Multimedia feature combining numerical simulations, surfing footage, and an accessible explanation of the study.

Featured in Surfline and Cliff Kapono's "Cliff Notes" video — Aug. 2026

My JFM research on the shoaling and overturning of waves was featured in a surfing-focused science communication video that received more than one million views on Instagram.

Interview on *AirTalk* (LAist 89.3 FM / NPR) — Jul. 2026

Discussed Hurricane Genevieve generated swell, coastal wave dynamics, and high-surf hazards in Southern California.

Interview with *CBS 8 San Diego* — Jun. 2026

Discussed a large south swell affecting Southern California, including hazardous surf, rip currents, and beach safety.

Featured in *El Surf en California*, Surf Channel TV documentary — Nov. 2025

Discussed research on the physics of breaking waves and highlighted notable surf breaks in the San Diego region.

Featured in Scripps Institution of Oceanography News — Jun. 2024

Profiled in "Time to Celebrate the Graduating Class of 2024" for undergraduate research on coastal processes, wave breaking, and development of the Wavedrifter and BathyBoard.

Featured in Waves (Brazil) article — Aug. 2023

"Dispositivo desvenda mecanica das ondas". International coverage of research on the Wavedrifter, a novel instrument developed for measuring the dynamics of breaking waves.

Featured in SurferToday article — Aug. 2023

"Wavedrifter: The Device That Decodes the Breaking of a Wave". Highlighted research on the development and deployment of the Wavedrifter, a novel instrument

for measuring wave steepening, overturning, and the transition to turbulence.

Member

American Geophysical Union 2026 – Present
American Society of Civil Engineers 2026 – Present

Skills

Numerical Models
OpenFOAM, fully nonlinear potential flow (FNPF) wave models
Programming & Computing
MATLAB, Python, LaTeX, Unix/Linux, Git, cluster computing
Field & Data Analysis
Emlid Studio PPK GNSS, single-beam echo sounders, ADCPs, pressure sensors, LiDAR, scientific scuba diving, and ATV operation

Field & Lab Experience

SHORECAL Experiment Jul. 2026
Oahu, Hawaii
Prepared and deployed ADVs using scientific diving and boat operations off White Plains Beach.

Functional Surfing Reef Laboratory and Field Campaigns 2024 – Present
San Diego, California and Ensenada, Mexico
Conducted laboratory experiments on small-scale artificial surfing reefs in the glass channel wave flume at the Scripps Institution of Oceanography Hydraulics Laboratory. Performed field deployment and recovery testing of reef anchoring systems at Scripps Beach, followed by full-scale reef deployments and monitoring in Todos Santos Bay, Baja California, Mexico.

BathyBoard Surf Zone Bathymetric Surveys 2023 – Present
Oahu, Hawaii, and San Diego, California
Planned and conducted GNSS and single-beam echo sounder surveys from a surfboard platform in energetic surf zone environments; supported equipment preparation, deployment, and recovery.

Rocky shOrelines eXperiment SImulation (ROXSI) 2022 – 2026
Monterey, California, and St. Thomas, U.S. Virgin Islands
Calibrated, programmed, deployed, and recovered ADCPs and pressure sensors using scientific diving and boat operations along rocky coastlines.

Long-Term Beach Profile Monitoring 2021 – 2024
San Diego, California
Conducted long-term monitoring of beach morphology using GNSS equipped ATVs, truck mounted LiDAR, and GNSS push dollies to collect high-resolution topobathymetric surveys of Southern California beaches. Participated in repeated surveys supporting investigations of shoreline change.

Wavedrifter Field Campaigns 2021 – 2023
California, Hawaii, and Tahiti
Deployed and recovered Lagrangian drifters in breaking waves to measure wave steepening, overturning, and the transition to turbulence.